

MULTI-SOURCE CANDIDATE DATA TRANSFORMER

Technical Design & Architecture Document | Step 1 Abstract

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Target Role: Engineering Intern (Jul-Dec 2026)

Language:

Node.js (ES Modules)

1. PIPELINE & INGESTION FLOW

The transformer executes as a deterministic, multi-stage batch processing pipeline:

- **Extract:** Parses structured CSVs, semi-structured ATS records, and unstructured PDF resumes.
- **Normalize:** Sanitizes data formats: lowercases emails; normalizes phones to E.164; maps dates to YYYY-MM; cleans skills.
- **Merge:** Groups candidates across files using email/phone hashes. Resolves property conflicts using source trust weights.
- **Validate:** Checks outputs against schema rules (types, ISO country codes, and E.164 phone formats).
- **Project:** Applies dynamic mappings from config.json to select fields, remap keys, and handle missing value rules.

2. PDF HYPERLINK & SECTION EXTRACTION

To address unstructured resumes, the extraction engine employs customized text and binary parsers:

- **Section Segmentation:** Categorizes text blocks (Experience, Projects, Education, Skills) using regular expressions and fallbacks.
- **Annotation Link Parser:** Scans the PDF binary stream to extract embedded hyperlinks (LinkedIn, HackerRank, LeetCode, certificates) directly.
- **Project Link Mapping:** Scores repository URLs against project titles using a token-matching algorithm, pairing repos to projects with zero false-positives.

3. SCHEMAS & RELATION-TABLE SHAPING

Outputs strictly align with the default schema defined in the technical specification:

- **Headline at Root:** Headline is formatted as a root-level string located right after the location and links keys.
- **Isolated Projects:** Personal projects are extracted into a root-level key (projects: [{ name, summary, github_link }]) separate from experience.
- **Certifications Key:** Certifications are parsed into a root-level array and mapped to their respective Google Drive credentials in order.
- **2D Arrays (Tables):** Skills and Provenance are output as arrays of arrays (e.g. skills: [[name, confidence, sources[]]]) representing tables.

4. MERGING & CONFLICT RESOLUTION

We match profiles across multiple sources. Conflicting fields are resolved using a confidence matrix:

- **Trust Weights:** Recruiter CSV holds higher trust for admin data (Name, email: 0.95). Unstructured PDF resumes are trusted for skills, experience, projects, and education (0.90).
- **Core Merges:** Deduplicates contact info. Cross-references current company titles. Unions skill catalogs.
- **Confidence Scores:** Skills get +0.05 confidence if verified in experience logs. Overall profile confidence is a weighted average of populated fields.
- **Provenance:** Every merged property logs its origin (file name and parser method) for auditable traceability.

5. RUNTIME SCHEMA PROJECTION

A schema-agnostic projection layer reads config.json to reshape candidate outputs dynamically:

- **Path Projection:** Supports selectors like emails[0], location.city, and 2D arrays (skills[].name).
- **Renaming & Toggles:** Remaps paths to custom target keys; toggles provenance and confidence metadata output.
- **Missing Policies:** If a path is empty, it can be set to null, omitted from result.json, or throw a validation error.

6. EDGE CASES & SCOPE DEFERRALS

- **Formatting Collisions:** Low-quality inputs (spaces, parentheses) are E.164-normalized. For missing country codes, we cross-reference candidate region mapping.
- **Date Ambiguity:** Dates (June 2021, 06/2021, present) are parsed using a token-regex match. Missing months default to 01.
- **Link False-Positives:** Tokenizes and scores repository URLs against project titles, preventing mismatched project repo links.
- **Time-Pressure Deferral:** Deferred OCR image resume scanning and live API connection verification in favor of offline binary parsing to ensure 100% reliability.